

ai-os-new

Type / Contract Re-Audit Report

Verification audit after bug fixes. Original: 23 findings. Remaining: 23 findings.

Severity	Count
CRITICAL	4
HIGH	7
MEDIUM	9
LOW	3
Total	23

TC-01	CRITICAL	EventMap wildcard index signature bypasses all type safety
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Files: event-bus.ts

Problem: EventMap declares [event:string]:unknown which makes typed map a suggestion not enforcement. Any string key valid. 40+ events not in EventMap still compile.

Fix: Remove index signature. Add all missing events. Use Record<never,never> as catch-all.

TC-02	CRITICAL	memory:updated payload type entirely wrong across definitions
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Files: event-bus.ts, domain-events.ts, domain-types.ts, memory-engine.ts

Problem: EventMap/DomainEventMap say {collection,action,id?} but actual emission is MemoryEntry[]. EventPayloads/Validators correctly say MemoryEntry[]. Consumers crash accessing .collection on array.

Fix: Change EventMap and DomainEventMap memory:updated type to MemoryEntry[].

TC-03	CRITICAL	keystate:updated/removed emitted payload doesn't match declared type
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Files: key-state-store.ts, domain-events.ts

Problem: DomainEventMap says {keyId,provider,state} but KeyStateStore emits {id,state}. Neither event in EventMap. Consumers access payload.keyId get undefined.

Fix: Align KeyStateStore emission with DomainEventMap or vice versa. Add to EventMap.

TC-04	CRITICAL	Dual/conflicting DebatePhase enum definitions
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Files: debate-runtime.ts, debate-state.ts

Problem: debate-runtime.ts: created|queued|initializing|active|... debate-state.ts: pending|opening|argumentation|rebuttal|... Only 'consensus' overlaps. Incompatible state machines.

Fix: Consolidate to single DebatePhase type. Deprecate debate-state.ts version.

TC-05	HIGH	debate:consensus payload type doesn't match actual emission
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Files: event-bus.ts, domain-events.ts, debate-service.ts

Problem: EventMap says {sessionId,confidence,claims[]}. DomainEventMap says unknown. Validators say {topic,consensus,convergenceScore}. Actual is {topic,consensus,convergenceScore,synthesis?}. Three different shapes.

Fix: Align all three to {topic,consensus,convergenceScore,synthesis?}.

TC-06	HIGH	settings:updated payload mismatches across definitions
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Files: event-bus.ts, domain-events.ts, settings-service.ts

Problem: EventMap says {settings,changes}. DomainEventMap says {key}. Actual is {settings:{...},changes:validated}. DomainEventMap completely wrong.

Fix: Update DomainEventMap to {settings:SystemSettings,changes:Partial<SystemSettings>}.

TC-07	HIGH	skills:updated, mcp:updated, tools:updated array vs object confusion
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Files: event-bus.ts, domain-events.ts

Problem: DomainEventMap declares {action,skillId?} objects. EventMap declares arrays. Incompatible shapes for same events.

Fix: Determine actual emission shapes and align all three type sources.

TC-08	HIGH	role:assigned/unassigned field name mismatch agentId vs nodeId
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Files: event-bus.ts, domain-events.ts, schema-types.ts

Problem: EventMap/DomainEventMap say {roleId,agentId}. Validators say {roleId,nodeId}. Validator rejects correctly-typed payloads.

Fix: Unify to agentId. Update validator.

TC-09	HIGH	agent:health:change type diverges across three sources
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Files: event-bus.ts, domain-events.ts, schema-types.ts

Problem: EventMap widens from/to to string. DomainEventMap uses AgentHealth enum. Validators use z.enum. EventMap loses enum constraint.

Fix: Use AgentHealth type in EventMap for from/to fields.

TC-10	HIGH	IEventBus interface erases all type information
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Files: interfaces.ts

Problem: on/emit use unknown for callback data and emit data. Any class implementing IEventBus loses typed EventMap guarantees.

Fix: Parameterize IEventBus with EventMap generic.

TC-11	HIGH	40+ events in EventValidators but not in EventMap
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Files: event-bus.ts, schema-types.ts

Problem: All DebateRuntime, Observability, and many domain events have validators but no EventMap entry. Work due to index signature. Zero type checking.

Fix: Add all events from EventValidators to EventMap with proper payload types.

TC-12	MEDIUM	system:node:removed field name mismatch nodeId vs id
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Files: domain-events.ts, schema-types.ts

Problem: DomainEventMap says {nodeId}. Validators say {id}. Inconsistent field names.

Fix: Align to one name. Prefer nodeId for domain consistency.

TC-13	MEDIUM	diagnostic:complete type completely different
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Files: event-bus.ts, domain-events.ts

Problem: EventMap: {id,scope,health,score,issueCount,timestamp}. DomainEventMap: {type,severity,summary}. No overlapping fields.

Fix: Align DomainEventMap to match EventMap/validator shape.

TC-14	MEDIUM	virtual:key:created payload diverges
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Files: event-bus.ts, domain-events.ts, schema-types.ts

Problem: EventMap: {virtualKey:unknown}. DomainEventMap: {virtualKeyId,provider,label}. EventValidators: {virtualKey:z.unknown()}.
Fix: Use DomainEventMap shape in EventMap.

TC-15	MEDIUM	CognitiveEventMap vs EventMap request:incoming/completed diverge
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Files: cognitive-events.ts, event-bus.ts

Problem: request:incoming and request:completed have completely different shapes across the two maps.

Fix: Determine actual emission and align both maps.

TC-16	MEDIUM	CacheService doesn't implement ICacheService
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Files: contracts/cache.ts, cache-service.ts

Problem: get() return type differs. tryGet/trySet not implemented. No runtime contract enforcement.

Fix: Either implement interface or remove it. Update return types.

TC-17	MEDIUM	HealthService.checkKey returns undefined but interface says null
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Files: contracts/health.ts, health-service.ts

Problem: IHealthService declares null but implementation returns undefined. null !== undefined in strict mode.

Fix: Change implementation to return null explicitly or change interface.

TC-18	MEDIUM	Unvalidated JSON.parse in serialization paths
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Files: storage-adapter.ts, memory-engine.ts, debate-session-persistence.ts

Problem: StorageAdapter.get uses JSON.parse(raw) as T with no validation. Corrupted JSON causes crashes or injects unexpected data.

Fix: Wrap JSON.parse results in Zod schema validation or runtime type guards.

TC - 19	MEDIUM	PluginSDK.emit uses unsafe cast to bypass EventMap
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Files: PluginSDK.ts

Problem: EventBus.emit(event as keyof EventMap, data) - cast bypasses type checking. Any arbitrary string from plugin accepted.

Fix: Validate plugin events against EventValidators before emitting.

TC - 20	MEDIUM	EventBus.onSafe falls back to unsafe cast when no validator
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Files: event-bus.ts

Problem: onSafe<T> without validator does raw as T. 'Safe' method name implies validation but is just as unsafe.

Fix: Throw/warn when no validator exists. Or rename to make conditional safety explicit.

TC - 21	LOW	Zod schemas use z.any() bypassing validation
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Files: schema-types.ts

Problem: ApiKeySchema uses z.any() for stats and config. ChatSessionSchema for metadata. CognitiveTraceSchema for steps. RoleSchema for permissions.

Fix: Replace z.any() with proper schemas matching TypeScript interfaces.

TC - 22	LOW	Tool interface index signature allows unrestricted access
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Files: \lm\core\types.ts

Problem: Tool declares [key:string]:unknown. tool.anyProperty compiles and returns unknown. Defeats purpose of typing known fields.

Fix: Remove index signature. Use Record<string,unknown> for extensions.

TC - 23	LOW	cross-tab broadcastCompatibility uses any type
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Files: cross-tab-state.ts

Problem: broadcastCompatibility(type, keyId, data: any) bypasses TypeScript safety. Also fabricates incomplete state objects.

Fix: Type data as union type for circuit breaker and rate limit states.